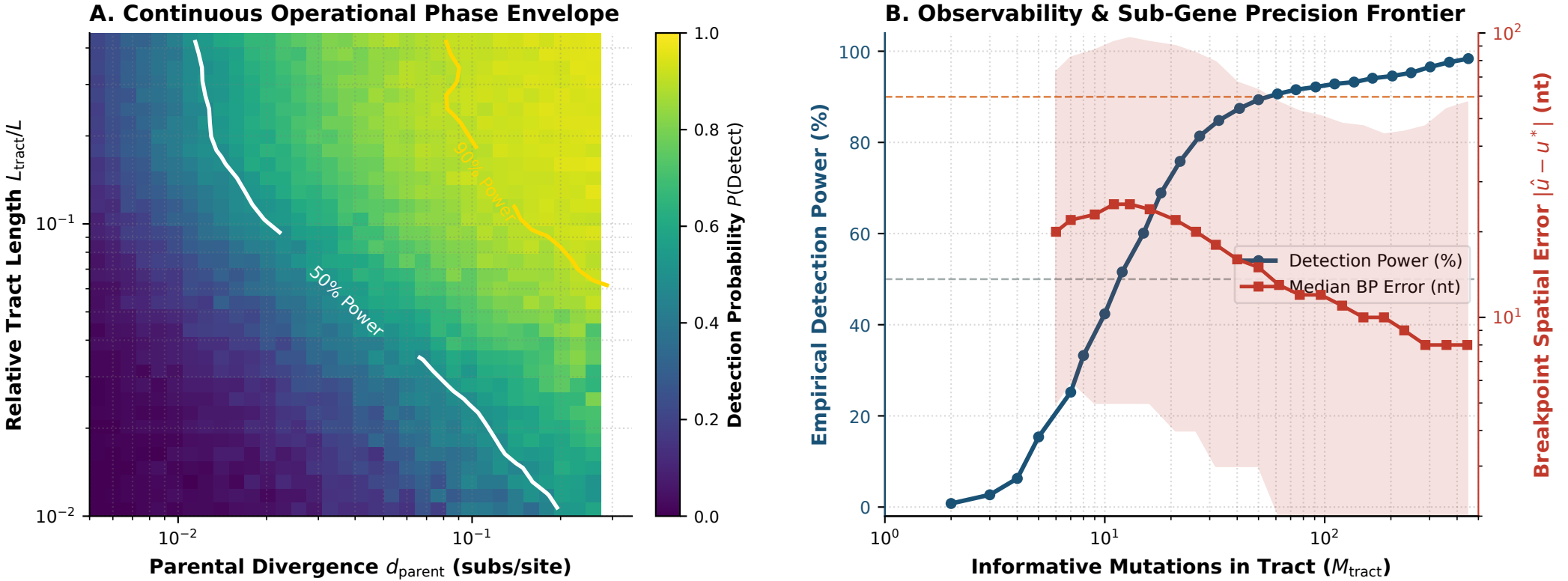

A. Continuous Operational Phase Envelope
Relative Tract Length L_{tract}/L
10^{-1}
10^{-2}
10^{-2}
10^{-1}
Parental Divergence d_{parent} (subs/site)
90% Power
50% Power
1.0
0.8
0.6
0.4
0.2
0.0
Detection Probability P(Detect)
B. Observability & Sub-Gene Precision Frontier
Empirical Detection Power (%)
100
80
60
40
20
0
10^0
10^1
10^2
Informative Mutations in Tract (M_{tract})
10^2
10^1
Breakpoint Spatial Error $|\hat{u} - u^*|$ (nt)
Detection Power (%)
Median BP Error (nt)